Ridge Regression

Ridge Regression will teach us how to handle cases where we have **many predictors**, especially when those predictors are correlated with each other. This connects directly to the Bias-Variance Trade-off because Ridge reduces model variance by shrinking coefficients.

Problem	Problem title	Research-paper anchor	Dataset needed
Problem 8	Predicting Energy Industry Excess Returns	Fama and French's five-factor asset-pricing model; Ridge methodology from Hoerl and Kennard	Kenneth French Data Library: Fama/French 5 Factors + 10 Industry Portfolios
Problem 9	Forecasting U.S. GDP Growth	Stock and Watson's macroeconomic forecasting with many predictors; FRED-MD dataset paper by McCracken and Ng	FRED-MD monthly macro dataset + FRED quarterly real GDP
Problem 10	Predicting CO_2 Emissions Intensity	STIRPAT / IPAT environmental-impact literature, especially York, Rosa and Dietz	Our World in Data CO_2 dataset + World Bank WDI indicators

Table 1: *Overview of Problems, Research Anchors, and Datasets*

8. Predicting Energy Industry Excess Returns Using Multiple Asset Pricing Factors

Research anchor: Fama, E.F. and French, K.R. (2015) 'A five-factor asset pricing model', *Journal of Financial Economics*, 116(1), pp. 1–22. DOI: 10.1016/j.jfineco.2014.10.010. The paper introduces the five-factor model using market, size, value, profitability, and investment factors.

Methodological anchor: Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge regression: Biased estimation for nonorthogonal problems', *Technometrics*, 12(1), pp. 55–67. DOI: 10.1080/00401706.1970.10488634. This is the original Ridge Regression paper.

Dataset: Kenneth R. French Data Library: Fama/French 5 Factors and 10 Industry Portfolios. The Kenneth French Data Library provides downloadable factor and portfolio return datasets widely used in empirical asset-pricing research.

In this problem, we will model **Energy industry excess returns** using the Fama-French five factors:

- Mkt-RF, SMB, HML, RMW, CMA

We will compare two models:

1. Ordinary Linear Regression (OLS)
2. Ridge Regression

The main question is: *Does Ridge Regression give a more stable model than ordinary Linear Regression when multiple asset-pricing factors are used?*

This problem is important because Ridge Regression introduces the idea of **regularisation**. Regularisation means we deliberately shrink coefficients to reduce overfitting and instability.

Findings

Ridge Regression did shrink the coefficients slightly, but it did **not** improve prediction performance compared with OLS in this particular problem.

Model	RMSE	MAE	Test R-squared
OLS Linear Regression	5.951606	4.363695	0.296767
Ridge Regression	5.953637	4.367643	0.296287

Table 2: *Performance Comparison for Energy Industry Excess Returns*

OLS performs very slightly better:

- OLS RMSE = 5.9516
- Ridge RMSE = 5.9536

But the difference is extremely small. Practically, both models perform almost the same.

The test $R^2$is around: 0.296

This means the model explains about **29.6% of the variation** in Energy industry excess returns in the test period.

Our selected Ridge penalty was: Best alpha = 3.7276

This means RidgeCV selected a moderate amount of regularisation. It did not choose a very tiny alpha, so it did apply some shrinkage, but the shrinkage was not strong enough to materially improve test performance.

Factor	OLS Coefficient	Ridge Coefficient
Mkt-RF	4.135009	4.082238
SMB	-0.660600	-0.648264
HML	-0.290746	-0.277574
RMW	-0.300387	-0.305841
CMA	1.067171	1.024706

Table 3: *Coefficient Comparison for Energy Industry Excess Returns*

Most coefficients became slightly smaller under Ridge. For example:

- Mkt-RF: 4.1350 → 4.0822
- SMB: -0.6606 → -0.6483
- HML: -0.2907 → -0.2776
- CMA: 1.0672 → 1.0247

The only coefficient that becomes slightly larger in absolute value is *RMW*, but overall, the Ridge model reduces the total size of the coefficient vector. This is normal because Ridge shrinks coefficients jointly, not necessarily every single coefficient individually.

Model	Residual Mean	Residual Standard Deviation
OLS Linear Regression	0.206087	5.957441
Ridge Regression	0.207153	5.959439

Table 4: *Residual Comparison for Energy Industry Excess Returns*

The residual standard deviation is almost identical for both models.

- OLS: 5.9574
- Ridge: 5.9594

So, Ridge did not reduce prediction error in this particular case.

Figure 1 (See Appendix) compares the coefficients estimated by OLS Linear Regression and Ridge Regression. The coefficients from both models are very similar, indicating that Ridge applies only mild shrinkage in this problem. The market factor, Mkt-RF, remains the dominant predictor in both models. Ridge slightly reduces the magnitude of most coefficients, which illustrates the regularization effect. However, the small difference between the OLS and Ridge coefficients suggests that the OLS model was already relatively stable with this limited set of five predictors.

Figure 2 (See Appendix) compares actual Energy industry excess returns with the predicted returns from OLS and Ridge Regression. The OLS and Ridge prediction lines almost overlap, showing that both models generate very similar forecasts. This is

consistent with the performance results, where RMSE, MAE, and test R-squared were almost identical for the two models.

The actual Energy excess return series is much more volatile than the predicted series. Both models capture some broad movements, but they fail to reproduce large extreme shocks, especially during crisis periods such as the global financial crisis and the 2020 energy-market shock. This suggests that **while the Fama-French factors explain part of Energy return variation, they do not fully capture sector-specific shocks.**

Overall, the plot supports the conclusion that Ridge Regression does not materially improve prediction in this problem. Its main contribution is coefficient shrinkage rather than better forecasting performance.

Together, the two figures show that Ridge Regression behaves as expected by shrinking coefficients slightly, but the shrinkage does not lead to a meaningful improvement in prediction. The OLS and Ridge models produce very similar coefficients, predictions, and errors. In this problem, Ridge serves mainly as a demonstration of regularization rather than as a clearly superior forecasting model.

Interpretation of Results

This problem compares Ordinary Least Squares Linear Regression with Ridge Regression for predicting Energy industry excess returns using the Fama-French five factors: Mkt-RF, SMB, HML, RMW, and CMA.

The results show that both models perform almost identically on the test set. The OLS model has an RMSE of 5.9516, while the Ridge model has an RMSE of 5.9536. The MAE is also slightly lower for OLS, at 4.3637 compared with 4.3676 for Ridge. The test R-squared values are also nearly 4.3676 for Ridge. The test R-squared identical, with OLS achieving 0.2968 and Ridge achieving 0.2963.

The selected Ridge alpha is 3.7276, indicating that RidgeCV applied a moderate amount of regularization. The coefficient table shows that Ridge shrinks most of the factor coefficients slightly toward zero. For example, the Mkt-RF coefficient falls from 4.1350 under OLS to 4.0822 under Ridge, and the CMA coefficient falls from 1.0672 to 1.0247. This demonstrates the main idea of Ridge Regression: it reduces coefficient size in order to make the model less sensitive to noise and instability.

However, in this particular problem, Ridge does not improve out-of-sample prediction performance. The OLS model performs marginally better than Ridge in terms of RMSE, MAE, and test R-squared. The difference is very small, so the practical conclusion is that both models perform almost the same.

This result is not a failure of Ridge Regression. Instead, it shows that Ridge is most useful when there are many predictors, strong multicollinearity, or high risk of overfitting. In this problem, there are only five asset-pricing factors, so the OLS model is already relatively stable. Ridge still demonstrates coefficient shrinkage, but it does not produce a meaningful forecasting improvement.

Overall, this exercise shows how Ridge Regression regularizes a linear model by shrinking coefficients. It also shows that regularization does not automatically improve prediction; its usefulness depends on the structure of the data and the degree of model instability.

Conclusion

Problem 8 introduces Ridge Regression as a regularized version of Linear Regression. Using Energy industry excess returns and Fama-French five-factor predictors, the Ridge model slightly shrinks the coefficients compared with OLS. However, it does not improve test performance in this case. The main learning is that Ridge Regression is valuable for controlling coefficient instability, but its benefits are more likely to appear when the model contains many correlated predictors or when OLS is prone to overfitting.

9. Forecasting U.S. GDP Growth Using Many Macroeconomic Predictors

Research anchor 1 - Macroeconomic forecasting with many predictors: Stock, J.H. and Watson, M.W. (2002) 'Macroeconomic Forecasting Using Diffusion Indexes', *Journal of Business & Economic Statistics*, 20(2), pp. 147–162. DOI: 10.1198/073500102317351921. This paper is a suitable research anchor because it studies how large sets of macroeconomic predictors can be used to forecast key macroeconomic variables.

Research anchor 2 - FRED-QD macroeconomic database: McCracken, M.W. and Ng, S. (2021) 'FRED-QD: A Quarterly Database for Macroeconomic Research', *Federal Reserve Bank of St. Louis Review*, 103(1), pp. 1–44. The FRED-QD database is designed as a large quarterly macroeconomic dataset for empirical macroeconomic research and is publicly available through the St. Louis Fed.

Dataset source: Federal Reserve Bank of St. Louis - FRED-QD: Quarterly Database for Macroeconomic Research. The St. Louis Fed describes FRED-MD and FRED-QD as large macroeconomic databases designed for empirical analysis of "big data", updated through FRED and publicly accessible for replication work.

Methodological anchor: Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge Regression: Biased Estimation for Nonorthogonal Problems', *Technometrics*, 12(1), pp. 55–67. DOI: 10.1080/00401706.1970.10488634.

In this problem, I will forecast **next-quarter U.S. real GDP growth** using a large set of quarterly macroeconomic indicators from FRED-QD.

The main question is: *Can Ridge Regression improve GDP-growth forecasting when many macroeconomic predictors are used?*

This is a natural Ridge Regression problem because macroeconomic variables are often highly correlated. For example, employment, industrial production, income, consumption, interest rates, and credit indicators often move together. Ordinary Linear Regression can become unstable when many correlated predictors are included. Ridge Regression helps by shrinking coefficients and reducing instability.

- The **target** variable is: Next-quarter real GDP growth

- The **predictors** are: Many current-quarter macroeconomic indicators from FRED-QD

Our model will use current-quarter macroeconomic information to predict next-quarter GDP growth.

Findings

This is one of those cases where the model teaches us more through its failure than through a clean improvement.

Ridge Regression dramatically improves over unstable OLS, but it still does not beat the simple mean benchmark.

Model	RMSE	MAE	Test R-squared
Mean Benchmark	1.398856	0.640068	-0.063151
OLS Linear Regression	786.226676	672.359653	-335848.416534
Ridge Regression	1.534399	0.935797	-0.279162

Table 5: *Performance Result for US GDP Growth*

The OLS result is clearly unstable. Its RMSE is extremely large:

- OLS RMSE = 786.2267

This means the ordinary linear regression model completely failed out of sample. The very negative test R^2 confirms this.

Ridge performs much better than OLS:

- OLS RMSE = 786.2267
- Ridge RMSE = 1.5344

So Ridge successfully prevents the huge OLS instability. However, the mean benchmark is still better:

- Mean Benchmark RMSE = 1.3989

- Ridge RMSE = 1.5344

So, we should not claim that Ridge gives the best forecasting model. The correct conclusion is:

Ridge strongly improves over OLS, but it does not outperform a simple average-growth benchmark in this specification.

OLS failed badly because we used a very large number of macroeconomic predictors. Many of these variables are highly correlated with each other. For example, variables related to production, employment, consumption, prices, and interest rates often move together over the business cycle.

When many predictors are correlated, OLS can produce very large and unstable coefficients. That is exactly what happened here. Some OLS coefficients became extremely large:

- CPIMEDSL = -109.6652
- CPIAPPSL = -96.5669
- PCESVx = 87.0162

These large coefficients caused the OLS prediction line to explode in the actual-vs-predicted plot. This is a classic example of why Ridge Regression exists.

The selected Ridge alpha was:

- Best alpha = 1873.8174

This is a very large alpha, meaning Ridge applied strong regularisation. In simple terms: Ridge strongly shrank the coefficients to prevent the model from overreacting to the macroeconomic predictors.

This is visible in the coefficient table (See *Figure 3* in Appendix). The OLS coefficients are very large, while the Ridge coefficients are close to zero. This shows the core mechanism of Ridge Regression:

Ridge reduces coefficient instability by shrinking coefficients toward zero.

The coefficient plot (See *Figure 4* in Appendix) clearly shows the difference between OLS and Ridge. The OLS bars are very large, both positive and negative. This suggests that OLS

is giving excessive importance to some predictors. The Ridge bars are almost flat because Ridge has shrunk the coefficients dramatically. This means the Ridge model is much more conservative.

The actual-vs-predicted plot in *Figure 5* (See Appendix) makes the OLS failure very clear. The actual GDP growth line stays close to normal quarterly growth values. The Ridge prediction line also remains stable and close to the actual scale.

However, the OLS prediction line becomes extremely negative, reaching values below -1000. This is economically meaningless for quarterly GDP growth. This tells me:

The OLS model has **severely overfit the training data** and produces unrealistic out-of-sample predictions. **Ridge avoids this explosion.** Even though Ridge does not beat the mean benchmark, it produces reasonable-scale predictions, unlike OLS.

Interpretation

In this problem, I used a large set of quarterly macroeconomic predictors from FRED-QD to forecast next-quarter U.S. real GDP growth. I compared ordinary OLS Linear Regression with Ridge Regression and also included a simple mean benchmark.

The results show that OLS performs extremely poorly out of sample. The OLS model has an RMSE of 786.2267 and a test R-squared of -335848.4165. These values indicate that the **OLS model is highly unstable** and produces unrealistic predictions. This is likely because the model includes many macroeconomic predictors that are **strongly correlated** with each other. In such a high-dimensional and correlated setting, OLS can assign very large and unstable coefficients.

Ridge Regression performs much better than OLS. Its RMSE is 1.5344, compared with 786.2267 for OLS. This shows that Ridge successfully prevents the extreme instability observed in the OLS model. The selected Ridge alpha is 1873.8174, indicating strong regularisation. This means the model applies a **large penalty to coefficient size** and shrinks the coefficients substantially.

The coefficient comparison confirms this. OLS assigns very large coefficients to several predictors, while Ridge shrinks these coefficients close to zero. This demonstrates the main idea of Ridge Regression: it reduces coefficient instability by shrinking coefficients when predictors are numerous or highly correlated.

However, Ridge does not outperform the simple mean benchmark. The mean benchmark has an RMSE of 1.3989, which is lower than the Ridge RMSE of 1.5344. This means that, although Ridge is much more stable than OLS, it does not produce the best forecast in this specification.

The actual-vs-predicted plot supports this interpretation. The OLS prediction line becomes extremely negative and economically unrealistic, while the Ridge prediction line remains stable and close to the scale of actual GDP growth. This shows that **Ridge controls the overfitting problem,** but it does not necessarily guarantee superior forecasting performance.

Overall, this problem shows why Ridge Regression is useful in macroeconomic forecasting with many predictors. Ridge does not automatically beat simple benchmarks, but it can prevent the severe coefficient instability that appears in ordinary OLS when many correlated macroeconomic variables are included.

Limitation

A limitation of this exercise is that I used the FRED-QD variables in a simplified modelling pipeline rather than fully applying the official transformation codes for each macroeconomic series. So,, the results should be interpreted as a beginner-friendly Ridge Regression exercise rather than a fully research-grade macroeconomic forecasting replication. A more advanced version of this problem would apply the recommended FRED-QD transformations before estimating OLS and Ridge models.

Conclusion

Problem 9 demonstrates the value and limitation of Ridge Regression in a macroeconomic forecasting setting. OLS becomes highly unstable when many correlated predictors are used, producing unrealistic forecasts and extremely poor test performance. Ridge Regression solves much of this instability by strongly shrinking the coefficients. However, Ridge still does not beat the simple mean benchmark in this specification. Hence, the main lesson is that Ridge is useful for controlling overfitting and coefficient instability, but regularisation alone does not guarantee strong forecasting performance.

10. Predicting CO_2 Emissions Intensity Using Economic and Energy Indicators

Research anchor - STIRPAT environmental-impact framework: York, R., Rosa, E.A. and Dietz, T. (2003) 'STIRPAT, IPAT and ImPACT: Analytic tools for unpacking the driving forces of environmental impacts', *Ecological Economics*, 46(3), pp. 351–365. This paper is the main research anchor because it develops the STIRPAT approach for studying environmental impacts using drivers such as population, affluence and technology.

Methodological anchor - Ridge Regression: Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge Regression: Biased Estimation for Nonorthogonal Problems', *Technometrics*, 12(1), pp. 55–67. This is the original Ridge Regression paper, and it is relevant because this problem uses many correlated economic, energy and emissions predictors.

Dataset source: Our World in Data, *CO_2 and Greenhouse Gas Emissions Dataset*. The OWID catalogue describes this as a complete CO_2 and greenhouse-gas emissions dataset containing key metrics maintained by Our World in Data. The GitHub repository provides the downloadable owid-co2-data.csv file and explains that the dataset is maintained by OWID and updated using sources including the Global Carbon Budget and related emissions datasets.

In this problem, I will use Ridge Regression to predict a country's **CO_2 emissions intensity**, measured as:

$$CO_2 \text{ emissions per unit of GDP}$$

This is a climate-economics problem because emissions intensity tells us how carbon-intensive an economy is relative to its economic output. The broad research question is:

Can economic, population, energy and fossil-fuel indicators help predict a country's CO_2 emissions intensity?

I will compare:

1. OLS Linear Regression
2. Ridge Regression
3. Simple mean benchmark

Again, many of the predictors are naturally correlated. For example, GDP, population, energy use, fossil-fuel emissions, and CO_2 per capita are connected to each other. OLS may become unstable when predictors are correlated, while Ridge Regression can shrink coefficients and control this instability.

Findings

Ridge Regression clearly outperforms both OLS and the mean benchmark for predicting next-year CO_2 emissions intensity.

Model	RMSE	MAE	Test R-squared
Mean Benchmark	0.2181	0.1835	-0.4541
OLS Linear Regression	0.3345	0.1574	-2.4217
Ridge Regression	0.1657	0.1340	0.1610

Table 6: *Model Performance for US GDP Growth*

The **OLS model performs poorly.** Its RMSE is 0.3345 and its test R-squared is -2.4217. A negative test R-squared means that the model performs worse than a simple baseline prediction. This suggests that **OLS is unstable** in this setting.

The Ridge Regression model performs much better. Its RMSE falls to 0.1657 and its MAE falls to 0.1340. More importantly, its test R-squared becomes positive at 0.1610. This means that Ridge explains around 16.1% of the out-of-sample variation in next-year CO_2 emissions intensity.

This is an important result because it shows that Ridge does not merely shrink coefficients; in this problem, it also **improves prediction performance.**

The mean benchmark predicts next-year CO_2 emissions intensity using the average value from the training sample. OLS performs worse than the benchmark in terms of RMSE and test R-squared. This confirms that OLS is not reliable in this setting.

Ridge performs better than both OLS and the benchmark. It has the lowest RMSE, the lowest MAE, and the only positive test R-squared. This means that Ridge is the best-performing model among the three approaches.

The key takeaway from this table is that Ridge Regression adds value beyond a simple average forecast.

The final selected Ridge alpha is: 14166.4327. The large alpha tells us that Ridge needed to apply a strong penalty to large coefficients in order to produce a stable model. This confirms that the model requires **strong regularisation**.

The coefficient comparison table in *Figure 8* (See Appendix) shows how OLS and Ridge estimate the predictor effects.

OLS assigns relatively large coefficients to several variables, especially Log_Population and Log_GDP. Ridge shrinks these coefficients dramatically toward zero. For example, the OLS coefficient for Log_Population is 0.8836, but the Ridge coefficient is only -0.0055. Similarly, the OLS coefficient for Log_GDP is -0.8636, while the Ridge coefficient is -0.0050. These large differences show that Ridge strongly reduces coefficient size.

This is not a weakness of Ridge. It is exactly what Ridge is designed to do. Ridge controls instability by preventing the model from relying too heavily on any one predictor.

Among the Ridge coefficients, Energy_per_GDP remains one of the larger numeric coefficients at 0.0581. This is economically sensible because a country that uses more energy per unit of GDP is likely to have higher CO_2 emissions intensity. However, I should interpret these coefficients as model weights rather than causal effects, because the variables are standardised and regularised.

Model	Residual Mean	Residual Standard Deviation
OLS Linear Regression	0.0755	0.3260
Ridge Regression	-0.1062	0.1272

Table 7: *Residual Comparison for US GDP Growth*

The OLS residual standard deviation is 0.3260, while the Ridge residual standard deviation is only 0.1272. This means that Ridge produces much more tightly distributed errors. In other words, its prediction mistakes are smaller and more stable.

The Ridge residual mean is -0.1062. Since residuals are calculated as actual minus predicted, a negative residual mean suggests that **Ridge slightly overpredicts CO_2 intensity on average.** However, the much smaller residual standard deviation shows that Ridge is still the more reliable model overall.

This table supports the earlier performance results: Ridge produces more stable and accurate predictions than OLS.

The coefficient plot in *Figure 6* (See Appendix) visually confirms the regularisation effect. The OLS bars are much larger, especially for Log_Population, Log_GDP and Oil_CO2_per_Capita. In contrast, the Ridge bars are much smaller and closer to zero.

This shows that OLS is more sensitive to the predictor variables and assigns large weights to some of them. Ridge is more conservative because it shrinks the coefficients.

The plot helps to see why Ridge performs better in this problem. By shrinking unstable coefficients, Ridge reduces the risk that the model overfits the training data and performs badly on unseen data.

The actual versus predicted plot in *Figure 7* (See Appendix) compares model predictions against the actual next-year CO_2 per GDP values. The dashed line represents perfect prediction. Points closer to this line indicate better predictions.

The Ridge predictions are more stable and concentrated. OLS produces more extreme predictions for some observations, especially where predicted values are far away from the perfect prediction line.

This visual result matches the performance table. Ridge has a lower RMSE and a lower residual standard deviation, so its predictions are less extreme and more reliable. OLS produces larger errors, which is consistent with its negative test R-squared.

The plot also shows that Ridge is not perfect. Some points are still far from the perfect prediction line. However, compared with OLS, Ridge clearly produces a more controlled prediction pattern.

Interpretation of Results

In this problem, I used economic, population, energy and fossil-fuel indicators to predict next-year CO_2 emissions intensity, measured as CO_2 emissions per unit of GDP. I compared OLS Linear Regression, Ridge Regression and a simple mean benchmark.

The results show that Ridge Regression performs best overall. The Ridge model has an RMSE of 0.1657 and an MAE of 0.1340, both lower than the OLS model. It also achieves a positive test R-squared of 0.1610, meaning that it explains around 16.1% of the out-of-sample variation in next-year CO_2 emissions intensity.

OLS performs poorly in comparison. Its RMSE is 0.3345 and its test R-squared is -2.4217, which means it performs much worse than the simple benchmark. This suggests that OLS is not well suited to this setting, most likely because many of the predictors are highly correlated. Variables such as GDP, population, energy use, CO_2 per capita and fossil-fuel emissions are naturally connected, which can make OLS coefficients unstable.

The selected Ridge alpha is 14166.4327. This is a large value, showing that the model requires strong regularisation. In practical terms, Ridge applies a strong penalty to large coefficients and shrinks them toward zero. This helps control the instability that appears in the OLS model.

The coefficient comparison shows this clearly. OLS assigns relatively large coefficients to variables such as Log_Population, Log_GDP and Oil_CO2_per_Capita. Ridge shrinks these coefficients substantially. For example, the OLS coefficient for Log_Population is 0.8836, while the Ridge coefficient is only -0.0055. Similarly, the OLS coefficient for Log_GDP is -0.8636, while the Ridge coefficient is -0.0050. This demonstrates the regularisation effect of Ridge Regression.

The coefficient plot confirms the same result visually. The OLS bars are large, while the Ridge bars are much smaller and closer to zero. This shows that Ridge produces a more conservative and stable model.

The actual versus predicted plot also supports the result. OLS produces more extreme predictions, while Ridge predictions are more controlled and closer to the scale of the actual CO_2 intensity values. Although Ridge does not perfectly predict emissions intensity, it produces smaller errors and performs better than both OLS and the mean benchmark.

Overall, this problem gives a clear example of where Ridge Regression is useful. In a climate-economics setting with many correlated economic and energy indicators, Ridge improves prediction and produces a more stable model than ordinary OLS.

Conclusion

Problem 10 shows that Ridge Regression can be useful in climate-economics prediction problems involving many correlated variables. OLS performs poorly and produces unstable coefficients, while Ridge shrinks coefficients and improves out-of-sample prediction. The Ridge model outperforms both OLS and the mean benchmark, making this the strongest Ridge Regression result in this phase so far. The main lesson is that regularisation is especially valuable when predictors are numerous, correlated and economically connected.

References

Fama, E.F. and French, K.R. (2015) 'A five-factor asset pricing model', *Journal of Financial Economics*, 116(1), pp. 1–22. doi: 10.1016/j.jfineco.2014.10.010.

Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge regression: Biased estimation for nonorthogonal problems', *Technometrics*, 12(1), pp. 55–67. doi: 10.1080/00401706.1970.10488634.

French, K.R. (2026) *Kenneth R. French Data Library*. Dartmouth College. Available at: Kenneth R. French Data Library.

Stock, J.H. and Watson, M.W. (2002) 'Macroeconomic forecasting using diffusion indexes', *Journal of Business & Economic Statistics*, 20(2), pp. 147–162. doi: 10.1198/073500102317351921.

McCracken, M.W. and Ng, S. (2021) 'FRED-QD: A quarterly database for macroeconomic research', *Federal Reserve Bank of St. Louis Review*, 103(1), pp. 1–44. Available at: Federal Reserve Bank of St. Louis Review.

Federal Reserve Bank of St. Louis (2026) *FRED-MD and FRED-QD: Monthly and Quarterly Databases for Macroeconomic Research*. Available at: Federal Reserve Bank of St. Louis.

Ritchie, H., Rosado, P. and Roser, M. (2023) 'CO_2 and Greenhouse Gas Emissions', *Our World in Data*. Available at: Our World in Data. Accessed: 19 July 2026.

Our World in Data (2026) *OWID CO_2 dataset*. Our World in Data Catalog. Available at: Our World in Data Catalog. Accessed: 19 July 2026.

Our World in Data (2026) *Data on CO_2 and greenhouse gas emissions by Our World in Data*. GitHub repository. Available at: OWID CO_2 Data Repository. Accessed: 19 July 2026.

York, R., Rosa, E.A. and Dietz, T. (2003) 'STIRPAT, IPAT and ImPACT: analytic tools for unpacking the driving forces of environmental impacts', *Ecological Economics*, 46(3), pp. 351–365. doi: 10.1016/S0921-8009(03)00188-5.

Appendix

A.8 Predicting Energy Industry Excess Returns Using Multiple Asset Pricing Factors

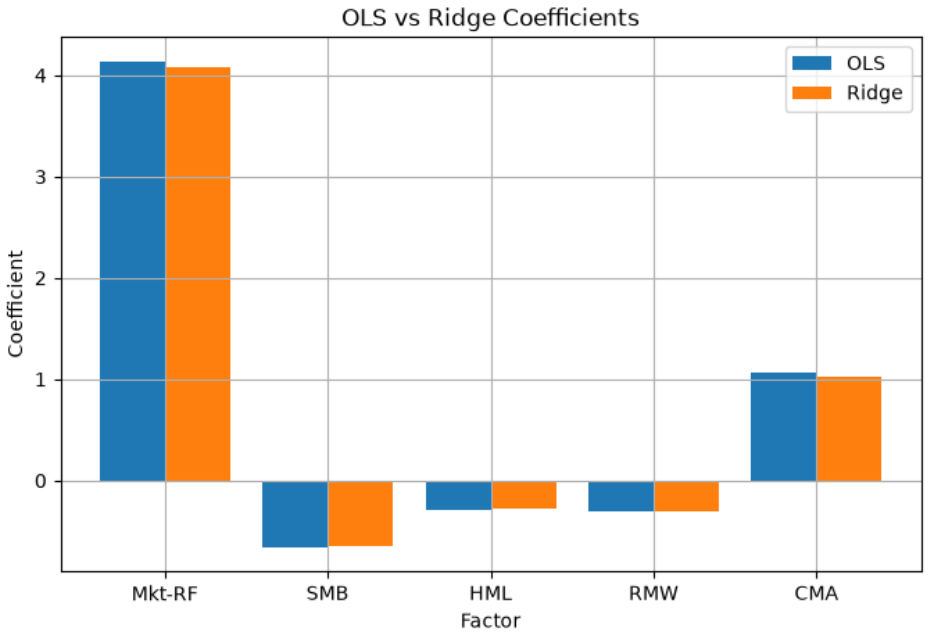


Figure 1: *OLS vs Ridge Coefficients*

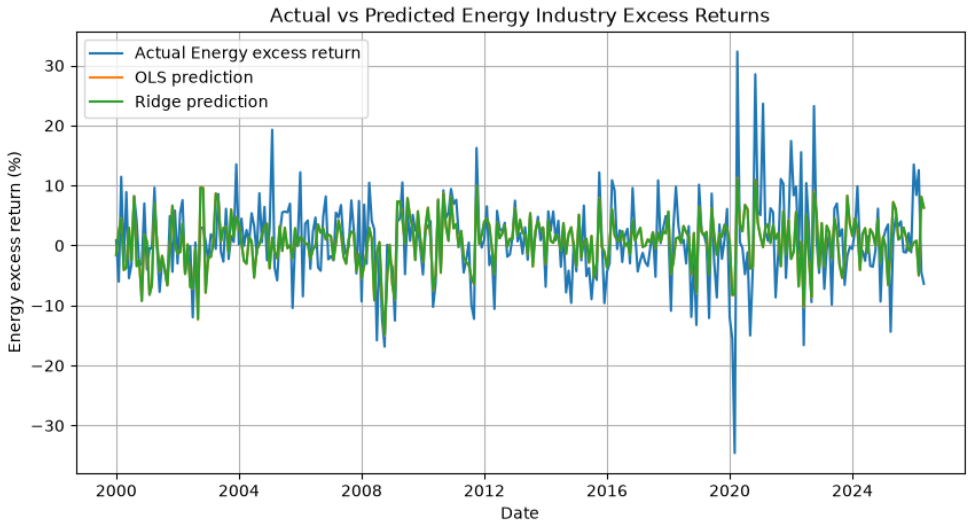


Figure 2: *Actual vs Predicted Energy Industry Excess Returns*

A.9 Forecasting U.S. GDP Growth Using Many Macroeconomic Predictors

	Feature	OLS Coefficient	Ridge Coefficient	Abs OLS Coefficient	Abs Ridge Coefficient
192	CPIMEDSL	-109.665157	-0.000890	109.665157	0.000890
190	CPIAPPSL	-96.566866	-0.000332	96.566866	0.000332
2	PCESVx	87.016202	-0.000902	87.016202	0.000902
114	DOTSRG3Q086SBEA	75.583077	-0.000524	75.583077	0.000524
198	CUSR0000SA0L5	73.021371	-0.000674	73.021371	0.000674
199	CES0600000008	63.829248	-0.000649	63.829248	0.000649
122	WPSID61	-63.256527	-0.000440	63.256527	0.000440
19	INDPRO	59.620895	-0.001238	59.620895	0.001238
208	CP3M	-59.180479	-0.009637	59.180479	0.009637
104	DFXARG3Q086SBEA	-59.089378	-0.000854	59.089378	0.000854
108	DHUTRG3Q086SBEA	-58.126992	-0.000548	58.126992	0.000548
29	IPB51220SQ	-54.338449	-0.001075	54.338449	0.001075
134	ULCNFB	53.584244	-0.000388	53.584244	0.000388
79	PERMIT	-48.890385	0.013841	48.890385	0.013841
155	NONREVSLx	-48.373055	-0.002772	48.373055	0.002772
156	REALLNx	47.876402	-0.002310	47.876402	0.002310
116	CPILFESL	45.448898	-0.000591	45.448898	0.000591
49	USTRADE	-44.989416	-0.000800	44.989416	0.000800
105	DCLORG3Q086SBEA	44.544876	-0.000608	44.544876	0.000608
71	HOABS	-44.212676	-0.000889	44.212676	0.000889

Figure 3: *Coefficients Table for US GDP Growth*

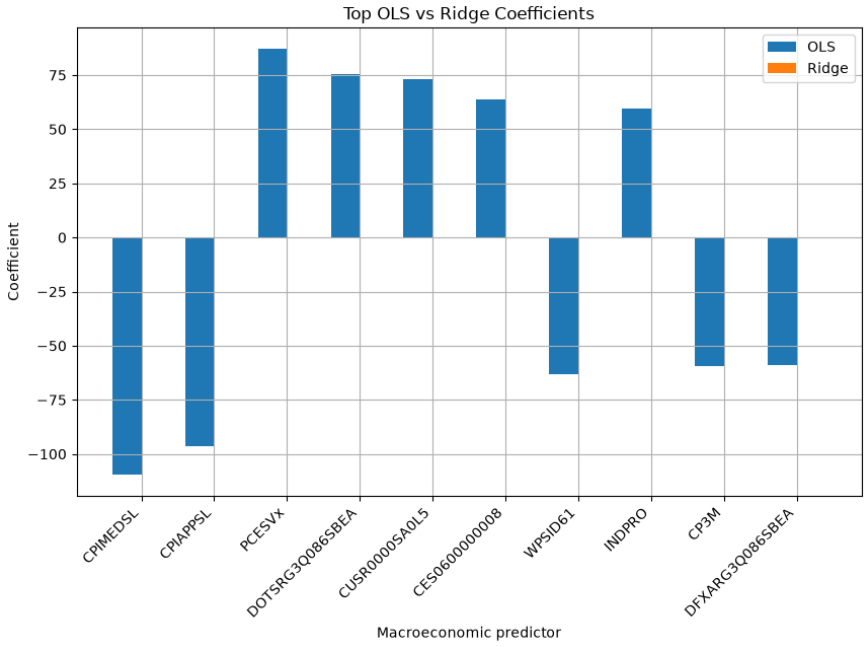


Figure 4: *Coefficients Plot for US GDP Growth*



Figure 5: *Actual vs Predicted US GDP Growth*

A.10 Predicting CO_2 Emissions Intensity Using Economic and Energy Indicators

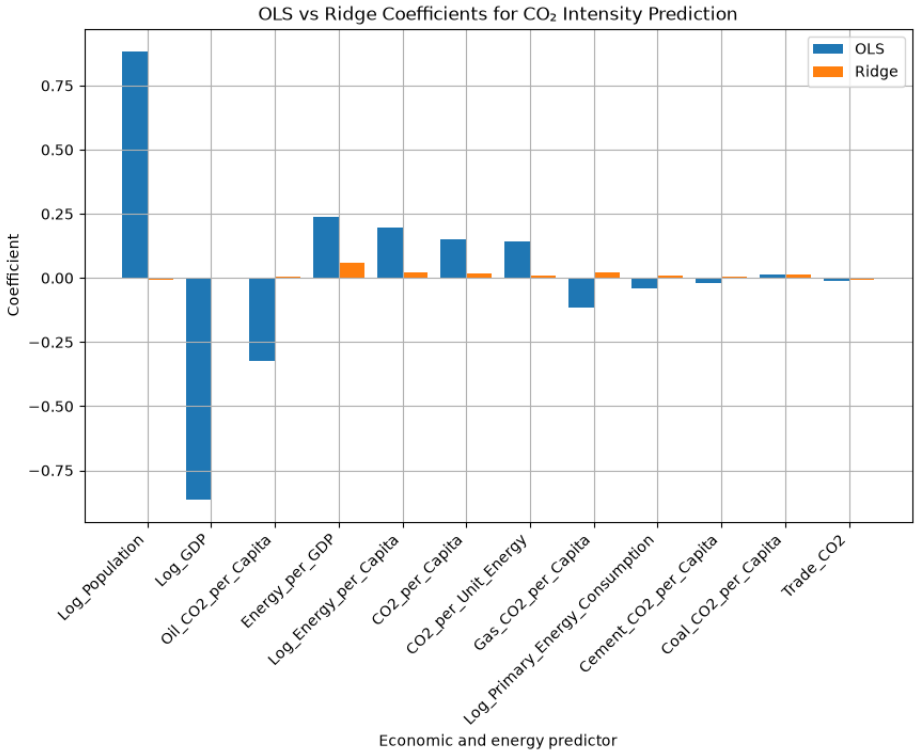


Figure 6: *OLS versus Ridge Coefficients for CO_2 Emissions Intensity Prediction*

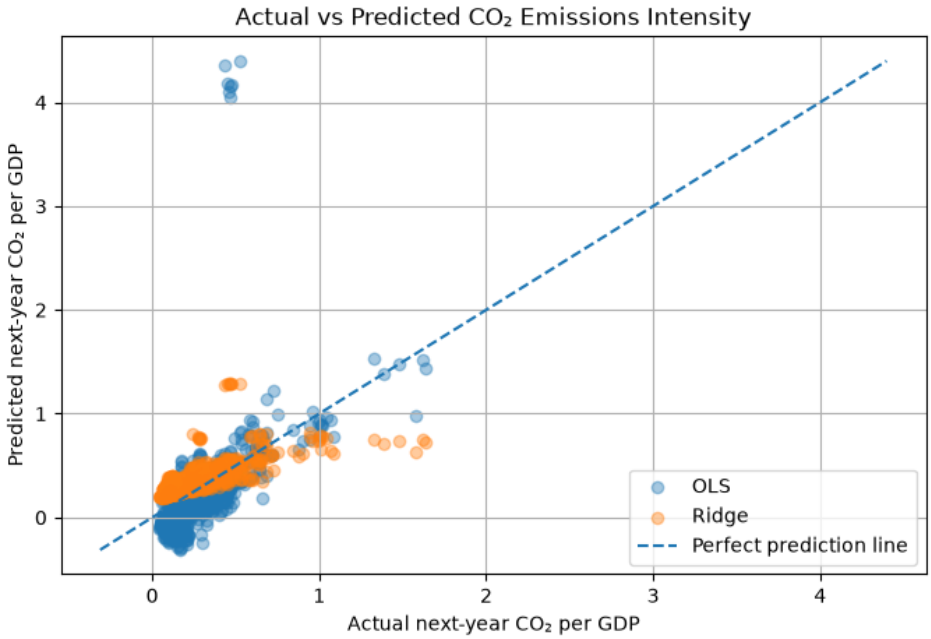


Figure 7: *Actual versus Predicted CO_2 Emissions Intensity*

	Feature	OLS Coefficient	Ridge Coefficient	Abs OLS Coefficient	Abs Ridge Coefficient
9	Log_Population	0.883628	-0.005512	0.883628	0.005512
8	Log_GDP	-0.863563	-0.005042	0.863563	0.005042
4	Oil_CO2_per_Capita	-0.321645	0.005176	0.321645	0.005176
1	Energy_per_GDP	0.236055	0.058126	0.236055	0.058126
11	Log_Energy_per_Capita	0.196898	0.019731	0.196898	0.019731
0	CO2_per_Capita	0.150910	0.018337	0.150910	0.018337
2	CO2_per_Unit_Energy	0.143294	0.009044	0.143294	0.009044
5	Gas_CO2_per_Capita	-0.114081	0.023301	0.114081	0.023301
10	Log_Primary_Energy_Consumption	-0.042440	0.010975	0.042440	0.010975
6	Cement_CO2_per_Capita	-0.020164	0.005323	0.020164	0.005323
3	Coal_CO2_per_Capita	0.014805	0.012552	0.014805	0.012552
7	Trade_CO2	-0.011612	-0.008397	0.011612	0.008397

Figure 8: *Coefficient Comparison Table for CO_2 Emissions Intensity*